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REGULAR ARTICLE

Which root architectural elements contribute the best to anchorage of *Pinus* species? Insights from *in silico* experiments

Ming Yang · Pauline Défossez · Frédéric Danjon · Sylvain Dupont · Thierry Fourcaud

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Abstract
Background and Aims Root anchorage function is crucial for tree survival as most trees are exposed to recurrent wind throughout their lifespan. Trees exhibit a large variability of root system architecture (RSA) due genetic and environmental factors. This study aims to understand the links between RSA and tree stability.
Methods A 3D biomechanical model was used to simulate tree overturning. To capture the variability of sinker RSA, fourteen virtual root patterns were created from an ensemble average of measured *Pinus pinaster* root systems. Root virtual patterns and tree-pulling simulations were verified against experimental data.
Results The model predicts realistic tree anchorage strength, root stress, and failure patterns. Only a few root components contribute significantly to anchorage strength. The taproot contributes the most to anchorage rigidity, representing 61 % of the anchorage strength. The windward roots failure drives ultimate anchorage failure, representing 25 % of the anchorage strength. Simulations show that root secondary thickening induces higher anchorage rigidity and increases anchorage strength by 58 %.
Conclusions This innovative approach appears promising for describing tree stability and its acclimation to external constraints.

Keywords Root system architecture · Numerical modelling · Root anchorage · Root stress · Failure patterns

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M. Yang · P. Défossez (✉) · S. Dupont
ISPA, INRA, Bordeaux Sciences Agro, F-33140 Villenave d'Ornon, France
e-mail: pauline.defossez@bordeaux.inra.fr

F. Danjon
BIOGECO, INRA, Université de Bordeaux, F-33610 Cestas, France

T. Fourcaud
AMAP UMR 0931, CIRAD, Boulevard de la Lironde, F-34398 Montpellier Cedex 5, France

Introduction

Anchorage is one of the primary functions of roots to guarantee plant survival. The anchorage function of a self-supporting plant is to hold its aerial parts upright and prevent its structure from falling or uprooting under gravity, air or water flows, floods, grazing herbivores, etc. (Schutten et al. 2005; Fournier et al. 2006; James et al. 2006; Crouzy et al. 2013; Starko et al. 2015). Like most terrestrial plants growing under natural conditions, trees are continuously exposed to winds throughout their lifespan. Tree response to wind loading is complex as it involves the architecture of both aerial and belowground parts, whose behaviour depends on their material and physical properties, and their interaction with the environmental media (air flow and soil; see Gardiner et al. 2016 for a review). The tree

crown plays a predominant role in the aerodynamic behaviour of aerial parts of the tree and is highly responsive to winds (Telewski and Jaffe 1986; Sellier et al. 2006; Rodriguez et al. 2008; Sellier et al. 2008; Sellier and Fourcaud 2009; Tadrist et al. 2014). Tree crown architecture is easier to define than root system architecture, the latter being hidden in the soil (Barthélémy and Caraglio 2007). For these reasons, the response of tree aerial parts to wind loads is better known than the root-soil system response. Although predicting tree resistance to uprooting is essential for evaluating the vulnerability of forests to extreme climate events (wind and snow storms, etc.), the mechanism of tree anchorage is still only partially described in risk prediction models (Gardiner et al. 2000; Dupont et al. 2015; Dupont 2016).

Root system architecture (RSA) is complex, varying among tree species and trees of the same species, depending on genetic and environmental factors, and changing throughout developmental stages (Köstler et al. 1968; Atger and Edelin 1994; Schroth 1998; Puhe 2003; Stokes et al. 2009; Ji et al. 2012). Above all, the development of RSA reflects the inherent characteristics of the tree species in accordance with their functional aspects, especially tree anchorage, water and nutrient acquisition. Four patterns of RSA have been identified among various tree species, namely the plate root system, the tap root system, the heart root system, and the sinker root system (Köstler et al. 1968). However this classification is too restrictive as for a given species; environmental factors may alter structural root development, which leads to a large variety of RSA (Köstler et al. 1968). For example, hardpan layers and the water table in soil limit the taproot development or the rooting depth (Coutts 1989; Ray and Nicoll 1998). Silvicultural practices and nursery handling (mechanical weeding, thinning, etc.) also influence local root growth (Fraser and Gardiner 1967; Ruel 1995; Danjon et al. 1999a; Puhe 2003). Fungal diseases may cause major root losses (*Heterobasidion annosum*, etc.; Fraser 1962). For some species, such as *Pinus pinaster*, trees are not able to regenerate an appropriate spatial arrangement of main structural roots from a defective juvenile root system for which some major root components are absent (Danjon et al. 2005). These modifications in RSA may weaken the tree anchorage function. Nevertheless, root growth may acclimate to environmental conditions to compensate for the defective root system and mitigate certain environmental perturbations, such as winds, sloping terrain, etc. (Coutts et al. 1999; Di Iorio et al. 2005; Khuder et al. 2007). For instance, under windy conditions, windward and leeward structural roots may exhibit increasing root number, diameter (i.e. secondary thickening) or length to enhance tree anchorage (Stokes et al. 1995). Edge trees develop a larger windward root sector than inner trees of the same stand in order to better resist wind loads (Cucchi et al. 2004). The vertical bending resistance of windward and leeward roots may also be strengthened through root cross-section shapes resembling I- or T-beams, as observed for *Picea sitchensis* (Nicoll and Ray 1996; Coutts et al. 1999). In addition, wind-induced adaptive responses of roots are also influenced by sloping terrain (Nicoll et al. 2006a; Danjon et al. 2013b).

The diversity in RSA appears important for tree anchorage function. The role of RSA in tree anchorage has been studied by tree-pulling experiments and root architecture analyses. Tree-pulling experiments investigate tree anchorage by mimicking the uprooting process (Coutts 1983; Coutts 1986; Nicoll et al. 2006b). This approach allows for tree anchorage strength to be estimated and for the response of the most active shallow parts of the root-soil system to be described. Root architecture analyses focus on the development and plastic response of structural roots due to winds based on measured root architecture characteristics (root spatial arrangement, cross-sectional area, branching patterns, biomass allocation by root type, etc.; Coutts and Lewis 1983; Nicoll and Ray 1996; Coutts et al. 1999; Mickovski and Ennos 2003; Reubens et al. 2007; Danquechin Dorval et al. 2016). It is desirable to link these root characteristics to the tree anchorage function. Both tree-pulling experiments and root architecture analyses have highlighted the role of the taproot, windward and leeward roots in tree stability (Coutts 1983; Coutts 1986; Ennos et al. 1993; Crook and Ennos 1996; Ray and Nicoll 1998; Coutts et al. 1999; Moore 2000; Danjon et al. 2005). Root architecture analyses also suggest the importance of the symmetry of the radial distribution of shallow roots (Coutts et al. 1999; Danjon et al. 2005). However, the full picture of RSA in relation to tree anchorage function remains elusive due to the difficulty in accessing the entire belowground root-soil system during tree uprooting, and due to the high natural variability of RSA and site conditions. Therefore, a modelling approach is necessary.

Numerical finite element models have been developed to describe in detail the belowground anchorage components and their behaviour during tree overturning (Dupuy et al. 2005; Dupuy et al. 2007; Fourcaud et al. 2008; Rahardjo et al. 2009). The belowground overturning process is driven by a bending load applied on the tree stem. The overall root-soil system failure is expected to characterise tree anchorage strength (critical turning moment). These models describe the RSA either with schematic representations or fully digitised real root systems (Dupuy et al. 2007; Fourcaud et al. 2008). The tree uprooting process has been analysed with respect to different theoretical patterns of root spatial arrangement (heart, tap, herringbone and plate-like RSA; Dupuy et al. 2005). However, the physical failure of the root system was absent in these models as roots were assumed to sustain any loads without breakage (Dupuy et al. 2005; Dupuy et al. 2007; Fourcaud et al. 2008). This limits the models ability to describe tree anchorage accurately. Recently, we have developed a novel anchorage model, named "RootAnchor" that accounts for the mechanism of sequential root breakage during the root-soil system failure (Yang et al. 2014). RootAnchor can consider various forms of RSA and different material properties for roots and soil. Parameterised with measured material and root architectural data, RootAnchor has yielded results that are in good agreement with a tree-pulling experiment. This initial application of RootAnchor was, however, limited to a specific root system of a young *P. pinaster* tree (Yang et al. 2014). This paper aims to analyse the impacts of contrasting RSA on the tree anchorage function using RootAnchor. In particular, we attempt to quantify the role played by the main structural roots during tree overturning. For this study, *P. pinaster* was selected as a model species because its RSA is representative of sinker RSA observed for *Pinus* species that grow in sandy podzols. In addition, root architecture data and tree-pulling experiments are available for *P. pinaster* to describe typical sinker RSA and tree anchorage. In this study, the degree of complexity of the RSA was limited to 3D schematic root patterns. To represent typical sinker RSA, the main root components and dimensions of root patterns were deduced from measured root architectural data. Using a complete description of real root systems would have led to results that would be specific to individual trees, as root systems are highly variable from one to another. Our approach consists of presenting the mechanism of tree overturning as a simulation by RootAnchor, based on the procedure to generate simplified RSAs. Different spatial arrangements of RSA were investigated with major root removal to quantify the impacts of root architectural components on the tree anchorage strength. Finally, the effects on tree anchorage of root acclimation to wind were examined.

Materials and methods

RootAnchor model

The response of the root-soil system to an overturning load applied on the tree stem was simulated with RootAnchor in order to mimic a tree-pulling experiment (Yang et al. 2014). A tree was pulled at a low and constant speed through a fictive quasi-horizontal cable attached to a winch. Since we were only interested in the belowground response of the tree, the stem was simply modelled as a 1.6 m long beam with rigid-body motions. A horizontal displacement of 1.2 m was imposed at the top of the stem in the direction of the prevailing wind.

Roots are slender structures. During tree overturning, structural roots undergo tensile, compressive or flexural bending loads depending on their location and fail if the ultimate strength is reached. In RootAnchor, the mechanical behaviour of individual root segments is modelled as a 3D Timoshenko beam (Timoshenko 1930). Roots are linearly elastic when subjected to loads within their ultimate strength. An elastic-damage constitutive law is used to describe the root material state in individual root segments (Yang et al. 2014). A scalar criterion defines the onset condition of damage based on axial strain in the root:

$$f = \sqrt{\frac{\varepsilon_{11}^{t}}{\varepsilon_{11}^{c}}(\varepsilon_{11})^2 + \left[\varepsilon_{11}^{t} - \frac{\left(\varepsilon_{11}^{t}\right)^2}{\varepsilon_{11}^{c}}\right]\varepsilon_{11}} > \varepsilon_{11}^{t} \qquad (1)$$

where ε_{11} is the axial strain component in the beam section and E the longitudinal modulus of elasticity, and

$$\varepsilon_{11}^{t} = \frac{\sigma_{11}^{t}}{E}, \qquad (2)$$

$$\varepsilon_{11}^{c} = \frac{\sigma_{11}^{c}}{E}, \qquad (3)$$

where ${\sigma_{11}}^{i}$ (for $i = t$ or c) is the ultimate strength of roots. The subscripts t and c refer to tensile and compressive strength, respectively. Note that f equals to zero for ε_{11}

ranging from 0 to $\varepsilon_{11}^t - \varepsilon_{11}^c$ (ε_{11}^t is usually greater than ε_{11}^c for wood materials). If $f > \varepsilon_{11}^t$, the following scalar damage evolution law is applied:

$$d = 1 - \frac{\varepsilon_{11}^t}{f} e^{-\frac{E \cdot \varepsilon_{11}^t \left(f - \varepsilon_{11}^t\right) L_c}{G_f}}, \quad (4)$$

where L_c is the length of the beam element and G_f the fracture energy of the root material. The damage variable d increases from 0 to 1 as root material stiffness degrades (Yang et al. 2014). Although the first damages are only related to tensile and compressive behaviour in roots, flexural damage may also occur when the tensile strain of the convex part or the compressive strain of the concave part of the bending root, reaches the onset condition (Eq. (1)). Table 1 summarises the root mechanical parameters used in the model. These data were obtained by a combination of published measurements with structural roots and greenwood of similar species (Stokes et al. 1997a; Khuder et al. 2007; Kretschmann 2010). More information can be found in Yang et al. (2014) regarding the constitutive law, the parameterization using experimental data, and the validation for structural root materials.

During tree overturning, soil undergoes large deformations and may slide locally along with the overturning root system. The soil material was assumed homogeneous in the model, initially linearly elastic and plastic under large shearing stress. The soil shear strength τ due to frictional resistance and cohesion between soil particles in contact is described by the Mohr-Coulomb failure criterion as follows:

$$\tau = \sigma \cdot \tan\varphi + c, \quad (5)$$

where σ is the stress normal to the failure plane, φ the internal friction angle, and c the cohesion of the soil. Table 1 summarises sandy podzol mechanical properties used in the model. More information about soil measurements can be found in Yang et al. (2014).

The soil computational domain was 10 m long, 10 m wide and 5 m deep. The root system is located at the centre of the soil domain, in a 6 m long, 6 m width and 1.25 m deep parallelepiped volume discretized with a refined mesh. Root-soil interaction is modelled as "embedded" connection. The displacements of root segments are computed by interpolation of the soil displacements in the immediate vicinity. Compared to a physical root-soil frictional interface, this simplification provides satisfactory results and copes with the numerical difficulties of modelling complex geometry and contact conditions of the root system within the soil (Yang et al. 2014). To optimize the model accuracy and computational performance, the mesh density for both the root pattern and the soil was optimized through a convergence study (Online Resources 1 and 2).

RootAnchor was developed in Abaqus 6.13 (http://www.3ds.com/products-services/simulia/portfolio/Abaqus/). A Fortran subroutine UMAT was used to define root elastic and damage behaviour (Yang et al. 2014).

Architecture of the root patterns

To capture the average RSA of an adult *P. pinaster* tree, we developed a method to generate virtual root patterns from a dataset of root systems architectures (Fig. 1). A reference root pattern (referred to as case 0) was designed based on the typology proposed by Danjon et al. (2005) for adult *P. pinaster* (Fig. 1a). First, the root pattern is divided into four sectors. The sectors are

Table 1 Material properties of roots and soil considered in RootAnchor: elastoplastic (associated with Mohr-Coulomb yield criterion) for soil and elastic-failure for roots

Roots		Soil	
Item	Value	Item	Value
Density (kg m^{-3})	421.4	Density (kg m^{-3})	1410
Young's modulus (GPa)	8	Young's modulus (MPa)	19.86
Shear modulus/Young's modulus (−)	0.0755	Poisson ratio (−)	0.33
Tensile strength (MPa)	43.2	Cohesion (kPa)	21.402
Compressive strength (MPa)	20.6	Friction angle (°)	14.62
Energy of fracture (J m^{-2})	209.4	Dilation angle (°)	0
Viscous parameter	0.000075		

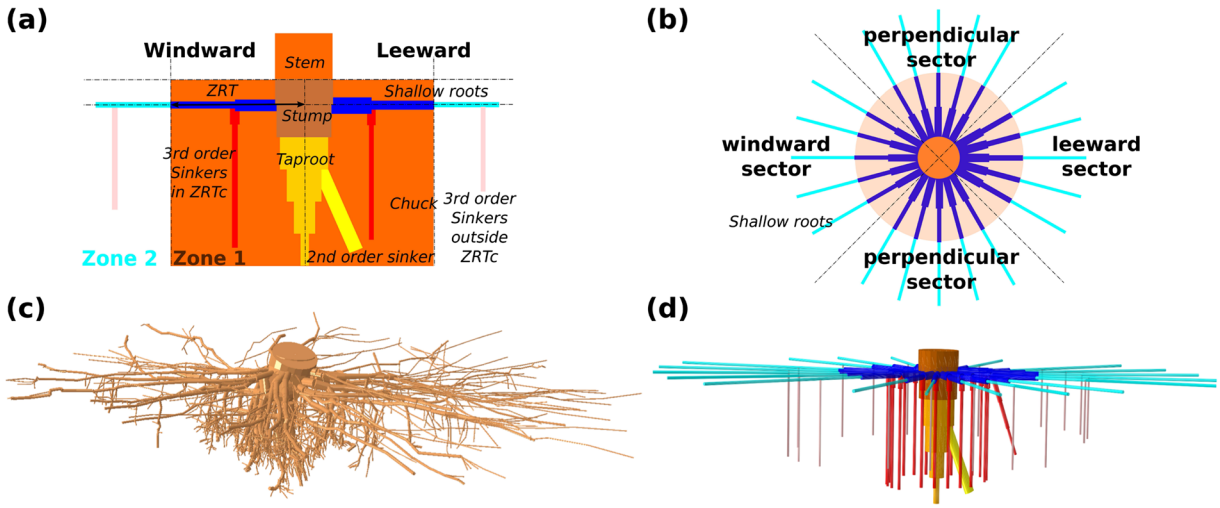


Fig. 1 Scheme of the methodology employed to generate the reference root pattern (case 0) of a *P. pinaster*. (**a**) Typology of the main root classes defined for adult *P. pinaster* (side view); (**b**) Typology of the main root classes defined for adult *P. pinaster* (top view); (**c**) View of a digitized root system from the L19 dataset; (**d**) View of the reference root pattern of case 0

centred at the stem base and defined following the prevailing wind direction referred to as 0°. Hereafter, the 0° ± 45° sector corresponds to the windward sector, the 180° ± 45° sector to the leeward sector, and the two other sectors to the perpendicular sectors (Fig. 1b). Second, the root pattern is divided in two zones, the cage and the zone outside the cage (respectively Zone 1 and Zone 2 in Fig. 1a). The cage is a cylinder region centred at the taproot. Its radius ZRT (Zone of Rapid Taper) corresponds to the mean length of the first shallow root segments that extend from the stem base with the most rapid taper. The depth of the cage corresponds to that of the taproot. Third, the root pattern is composed of main structural roots: the taproot, the shallow roots with respect to the prevailing wind direction, the second-order sinker roots carried by the taproot, the third-order sinker roots carried by the shallow roots within the cage, an unusually large leeward third-order root (termed as the chuck root for *P. pinaster*), and the third-order sinker roots outside the cage.

The dataset to parameterise the reference root pattern (case 0) was composed of seven measured root systems of 19-year-old trees with a mean Diameter at Breast Height (DBH) of 284.5 mm (Danjon et al. 2013a). This dataset is hereafter referred to as L19. Trees from L19 grew in the southwest of France (44°42′ N, 0°46′ W) in a sandy podzol with a discontinuous hardpan at about 0.7 m depth and a water table close to the soil surface during winter. The large structural roots were measured by segments of about 150 mm long. For each root segment, the spatial coordinates and the diameters of the segment extremities were recorded. Figure 1c illustrates a 3D digitized root system obtained from L19 where each root segment is represented by a cone. Further details about the root system measurement are available in Danjon et al. (2013a).

To build the reference root pattern from the L19 dataset, several assumptions were considered: (i) all roots were represented as straight beams with circular cross-sections without forks or root grafts; (ii) the taper of roots was only considered within the cage and was modelled by a chain of linked cylindrical root segments tapering off in diameter (see below); (iii) the average root volume per tree computed over L19 root systems was broadly conserved by the root pattern (Online Resources 3 and 4); (iv) the second-order sinker roots extending from the taproot were unified as one second-order sinker root on the leeward side, extending from the bottom of the stump; (v) all shallow roots were assumed horizontal while the taproot and third-order sinker roots were assumed vertical; (vi) other mid-depth horizontal roots, oblique roots and deep roots, were neglected as their relative biomass remained small.

With these assumptions, the geometry (lengths, diameters, tapers within the cage) and the number of each root component were deduced from the ensemble average of the seven L19 root systems (Table 2). The method used to estimate the root dimensions is illustrated for the taproot in Fig. 2. The taproot was modelled as a chain of four cylindrical segments and its volume was equal to the mean

Table 2 Main root components and their geometric characteristics used in the model were deduced from the ensemble average of L19 trees. They are compared with two groups of the measured root systems of the Bilos dataset, which include 12 intact specimens (Bilos Undamaged) and 12 damaged specimens during the Lothar and Martin windstorms (Bilos Uprooted). Note that all root geometric dimensions are in mm. Most root diameters of the Bilos dataset are quadratic mean values because two diameters were measured for each root section (major axis and minor axis)

Root items	Model case 0 Value based on L19	Bilos Undamaged Value	Bilos Uprooted Value	Bilos Undamaged Range	Bilos Uprooted Range
ZRT radius	870.3	1085.3	1134.4	856.9, 1316.1	913.6, 1674.5
Taproot basal diameter	292.2	241.3	249.6	94.7, 328.6	140.7, 487.4
Taproot distal diameter	51.1	22.0	15.6	1.0, 101.0	1.0, 88.6
Rooting depth (0 at the soil surface)	−1127.8	−1041.8	−1057.1	−893.0, −1273.0	−858.4, −1347.0
Windward sector					
Number of shallow roots	5	5	4	3, 7	0, 9
Basal diameter of shallow roots	48.2	107.3	108.1	67.8, 172.7	16.0, 219.1
Diameter of shallow roots beyond ZRT	22.0	19.7	14.6	8.8, 35.5	4.0, 31.5
Length of shallow roots	2442.8	2060.5	1875.7	1334.0, 2968.0	876.4, 2732.0
Number of sinkers in ZRT	5	7	6	3, 15	0, 16
Length of sinkers in ZRT	848.6	767.7	655.7	469.6, 1457.0	459.1, 821.9
Number of sinkers beyond ZRT	5	6	5	1, 18	0, 22
Length of sinkers beyond ZRT	620.1	615.4	552.4	320.9, 882.9	319.7, 921.0
Perpendicular sector					
Number of shallow roots	10	7	6	3, 9	1, 10
Basal diameter of shallow roots	59.1	123.2	122.4	84.8, 183.8	70.3, 197.1
Diameter of shallow roots beyond ZRT	22.6	21.7	21.5	11.0, 36.1	5.7, 58.2
Length of shallow roots	2115.1	2052.0	1944.8	1277.0, 2592.0	1513.0, 2607.0
Number of sinkers in ZRT	5	11	11	5, 20	1, 22
Length of sinkers in ZRT	814.9	714.3	630.0	462.6, 1112.0	425.0, 947.7
Number of sinkers beyond ZRT	5	6	6	0, 13	0, 12
Length of sinkers beyond ZRT	585.2	570.9	474.9	184.1, 880.9	112.7, 642.6
Leeward sector					
Number of shallow roots	5	2	2	0, 5	0, 5
Basal diameter of shallow roots	54.4	142.9	117.5	60.0, 209.6	18.0, 204.0
Diameter of shallow roots beyond ZRT	22.7	18.4	18.8	2.0, 26.8	7.0, 40.9
Length of shallow roots	2406.8	1916.3	2068.6	1033.0, 2438.0	1499.0, 2732.0
Length of the chuck	640.8	433.5	587.6	123.0, 665.2	197.2, 878.5
Number of sinkers in ZRT	5	4	6	0, 9	0, 9
Length of sinkers in ZRT	798.4	552.4	580.9	185.9, 890.9	178.9, 851.7
Number of sinkers beyond ZRT	5	3	3	0, 7	0, 10
Length of sinkers beyond ZRT	462.1	464.4	425.9	283.4, 551.8	205.4, 594.9

volume of L19 taproots. This method was applied to all root segments of case 0. As a few small roots were omitted in case 0 (i.e. deep roots and 2nd order oblique roots), the root volume of case 0 was smaller than the L19 mean root volume per tree (0.110 m^3 for case 0 against 0.137 m^3 for L19).

Figure 1d presents the root pattern of case 0 by comparison to the real root system of Fig. 1c. The root pattern

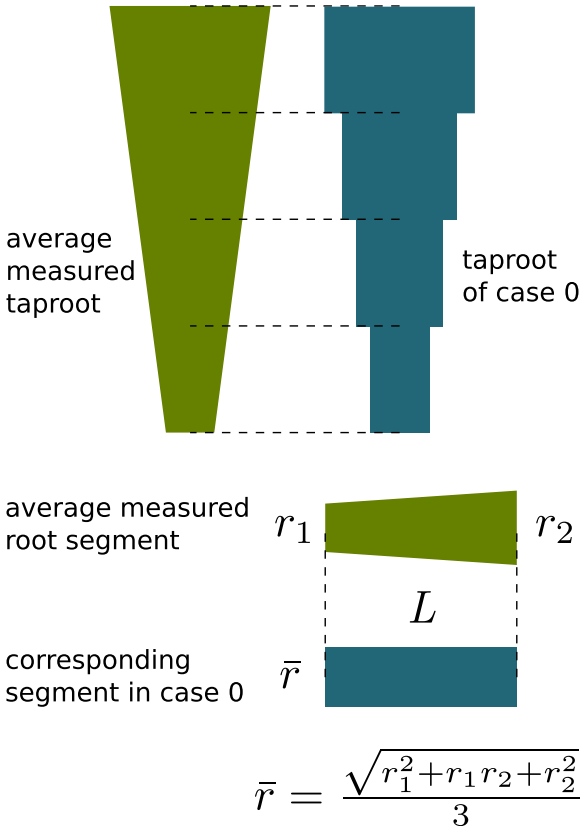


Fig. 2 Method used to generate the tapered root components by conserving the root volume. For example, the assumed conical taproot deduced from measurements, in *green*, is modelled by four cylinders, in *blue*, where each cylinder has a radius equal to the mean radius of the corresponding conical element of the measured taproot (radii r_1 and r_2)

represents well the root system asymmetry due to the selective biomass reinforcement in windward and leeward sectors found in the L19 dataset. The windward and leeward root volume represents 55 % of the total root volume of case 0. Windward and leeward shallow roots are on average 15 % longer than the perpendicular ones.

To account for the variability of RSA due to root losses, twelve root patterns were built by removing in each case (cases 1 to 12) one structural root component of the reference root pattern (case 0), all the other roots being unchanged. Hence, cases 1 to 12 do not account for acclimation mechanisms such as biomass reallocation or change in root shapes resulting from the loss of a root. The twelve root patterns are presented Fig. 3 and Table 3 describes the missing roots of each case.

To investigate the impact of wind-induced root acclimation on tree anchorage, we considered the example of windward and leeward root secondary thickening in a root system with a limited taproot development. This acclimation case is referred to as case p1 by comparison to case 1 with the same root pattern (limited taproot) but without root acclimation. To construct case p1, two assumptions were made based on experimental findings. First, in the absence of the taproot, the biomass was redistributed to the other major contributing components to tree anchorage, i.e. windward and leeward roots (Stokes et al. 1995). Second, for *P. pinaster*, this extra biomass was expressed as secondary thickening for all windward roots and for parts of leeward roots within the cage (Danjon et al. 2005), the root lengths being unchanged. To reallocate the taproot biomass to these roots, the ratio α between the acclimated and non-acclimated diameters was estimated as:

$$\alpha = \sqrt{\frac{V_{ref} + V_{\text{taproot}}}{V_{ref}}}, \tag{6}$$

where V_{ref} is the volume of the case 0 roots and $V_{taproot}$ the volume of the case 0 taproot. A 3D view of the root pattern of case p1 is given in Fig. 3.

A set of Python scripts was developed to automatically implement the fourteen root patterns in the RootAnchor model.

Data analysis

The relevance of the reference root pattern (case 0) and the diversity of RSA modelled by cases 1–12 and p1 are evaluated against another dataset of mature root systems of *P. pinaster* (see the Results section). This additional dataset is referred to as the Bilos dataset (Danjon et al. 2005), and corresponds to trees that grew in the same region as L19 trees (44°30′ N, 0°57′ W). It contains 24 trees of 50-year-old with a mean DBH of 378.4 mm. Twelve of these trees were uprooted by Lothar and Martin windstorms in 1999 and the other 12 were intact (Table 2). All the 24 root systems were carefully extracted without damaging the root segments. The same measurements as for L19 were done to characterize the RSA. Further details can be found in Online Resource 5 and in Danjon et al. (2005).

To evaluate quantitatively the tree-pulling simulations, we looked (1) at the turning moment (T_i, where the subscript i refers to the case number) as a function of the deflection angle (θ_i), both calculated at the stem base, where the initial slope of the $T_i(\theta_i)$ defines the rigidity of the root-soil system, and (2) at the critical turning moment (T_i^{crit}), which corresponds to the maximum value of T_i

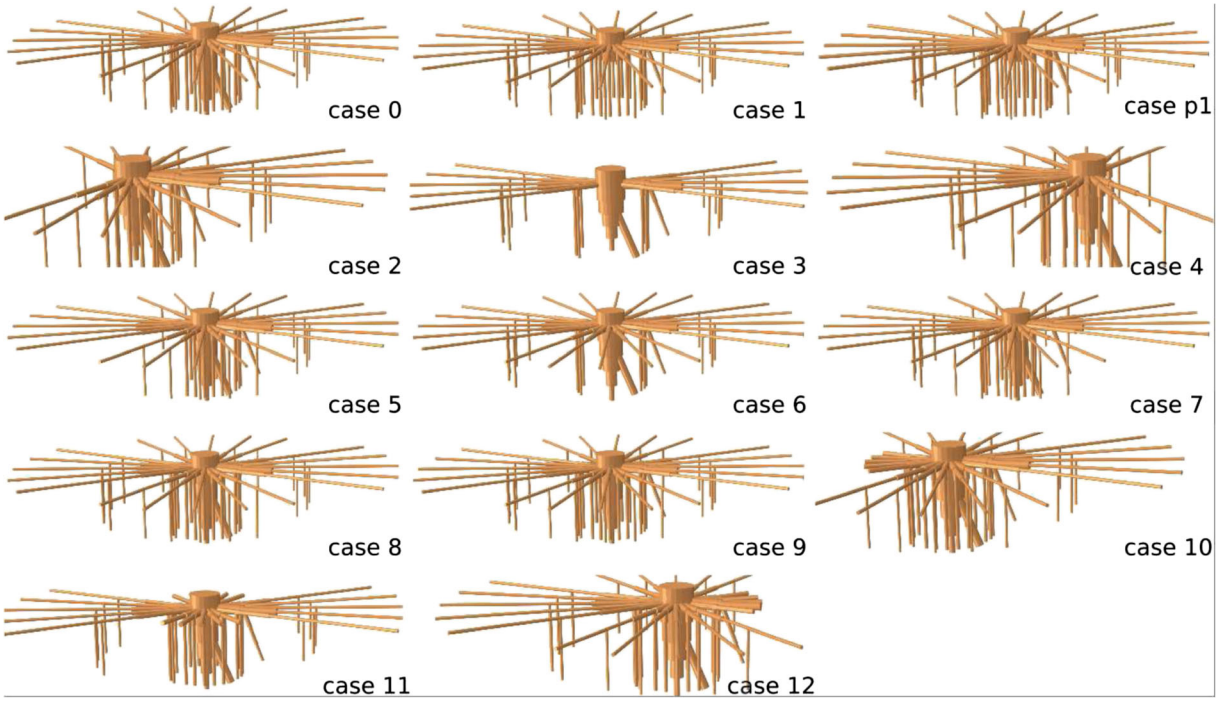


Fig. 3 View of the 14 root system architectures considered in this study. They are representative of contrasting architectures observed in natural conditions

and characterises the anchorage failure. The simulated T_0^{crit} was compared to the empirical estimation of the critical turning moment T_{Cucchi}^{crit} proposed by Cucchi et al. (2004). This estimation was obtained from a RootAnchor, and the mechanical contribution C_i defined by Eqs. (11) and (12). For case p1, the beneficial effect C^{plast} compared to case 1 is shown

Table 3 Description of the 14 root system patterns considered in this study (cases 0 to 12, and case p1). The occurrence of these root patterns in the Bilos dataset is indicated (the values in parentheses correspond to the occurrence for only uprooted trees of the Bilos dataset), as well as the critical turning moment (T_i^{crit}) predicted by

Cases (simulations)	Variation in RSA seen in the Bilos dataset	Occurrence in Bilos	T_i^{crit}(N.m)	C_i(%)
0	Reference case	-	36,240	-
1	Short taproot (≤350 mm)	3 (2)	14,195	60.8
2	Insufficient windward shallow roots (≤1)	3 (3)	27,263	24.8
3	Insufficient perpendicular shallow roots (≤3)	6 (5)	28,920	20.2
4	Insufficient leeward shallow roots (≤1)	9 (5)	33,757	6.9
5	Insufficient windward sinker roots (≤4)	4 (4)	32,041	11.6
6	Insufficient perpendicular sinker roots (≤8)	3 (1)	33,882	6.5
7	Insufficient leeward sinker roots (≤4)	6 (2)	36,662	−1.2
8	No chuck root	3 (2)	37,073	−2.3
9	No second order sinker roots	4 (3)	36,443	−0.6
10	Short windward shallow roots (≤350 mm outside the cage)	4 (2)	35,685	1.5
11	Short perpendicular shallow roots (≤350 mm outside the cage)	2 (1)	36,354	−0.3
12	Short leeward shallow roots (≤350 mm outside the cage)	1 (0)	35,436	2.2
p1	No taproot with selective reinforcement	-	22,369	57.6

campaign of 100 tree-pulling experiments performed on *P. pinaster* in the same region as L19 trees, covering a large range of tree sizes. T^{crit}_{Cucchi} was linearly correlated to $H \times DBH^2$, where H is the tree height. Three empirical expressions were obtained by Cucchi et al. (2004) depending on the site conditions:

$$T^{crit}_{Cucchi} = 35462 \times H \times DBH^2 + 2498.8 \left(r^2 = 0.88\right), \quad (7)$$

$$T^{crit}_{Cucchi} = 44848 \times H \times DBH^2 + 3033.0 \left(r^2 = 0.92\right), \quad (8)$$

$$T^{crit}_{Cucchi} = 39962 \times H \times DBH^2 + 7605.0 \left(r^2 = 0.90\right). \quad (9)$$

To estimate the empirical critical turning moment ($T^{crit}_{0,\ Cucchi}$) of our case 0 from Cucchi et al.'s empirical expressions, we estimated $H \times DBH^2$ from a linear correlation between $H \times DBH^2$ and the total root volume of each tree of L19, because the root volume of case 0 is smaller than the mean root volume of L19 trees ($volume_{root}$). This correlation was established from the L19 data set and verified for the Bilos dataset (Online resource 6):

$$H \times DBH^2 = 4.5564 \times volume_{root} + 0.7517 \left(r^2 = 0.56\right). \quad (10)$$

By applying this relationship to the reference root pattern for which the total root volume was 0.11 m^3, we obtained $H \times DBH^2 = 1.26\ m^3$. This leads to $T^{crit}_{0,\ Cucchi}$ ranging from 47.1 to 59.5 kN.m according to Eqs. (7)–(9).

To evaluate the mechanical contribution C_i of each structural root class and to quantify the influence of contrasting RSA, the critical turning moment T^{crit}_i from cases 1 to 12 and p1 were compared to that of case 0 (T^{crit}_0) as in Fourcaud et al. (2008):

$$C_i = \frac{T^{crit}_0 - T^{crit}_i}{T^{crit}_0} \times 100\%. \quad (11)$$

To examine how the adaptive plastic response of roots to the wind, i.e. the reallocation of biomass in windward and leeward roots, affects the critical turning moment, T^{crit}_{p1} was compared to T^{crit}_1:

$$C^{plast} = \frac{T^{crit}_{p1} - T^{crit}_1}{T^{crit}_1} \times 100\%, \quad (12)$$

where a positive value of C^{plast} quantifies an immediate beneficial effect resulting from selective reinforcement of root biomass.

Results

Verification of the root patterns

The root patterns constructed from the L19 dataset (cases 0 to 12) were first compared with those from Bilos dataset. Table 2 shows that case 0 exhibited similar architectural characteristics to Bilos trees in terms of the number of roots (in windward, leeward and perpendicular sectors) and root dimensions. For instance, the basal diameter of the taproot was 292.2 mm in case 0 against 249.6 mm for uprooted Bilos trees. The number of windward shallow roots is 5 in case 0 and 5 for undamaged Bilos trees. There were a few differences between Bilos and L19 trees considering the older age of Bilos trees. In particular, the mean DBH of Bilos trees was about 380 mm, against 280 mm for L19 trees; Bilos root systems exhibited an average ZRT (radius of the cage) 26 % larger than that of case 0; and the shallow roots were on average thicker at the root base of Bilos trees (Table 2). Furthermore, in Bilos trees, the volume of windward and leeward roots represented about 60 % of the root system volume, suggesting stronger selective reinforcement than in case 0, probably due to a longer acclimation of mature root systems to prevailing winds (Danjon et al. 2005). For comparison, the volume of windward and leeward roots represented 66 % of the total root volume in case p1, against 55 % in case 1, and about 60 % for Bilos trees. This illustrates a stronger asymmetry of the RSA induced by root adaptive plastic response to wind stimuli and taproot limitation. More detailed data are provided in Online Resource 7.

Regarding the representativity of the root patterns of cases 1 to 12, Table 3 shows the occurrence of RSA with root losses in the Bilos dataset similar to our cases 1–12. For all our cases, we found corresponding root system specimens in Bilos dataset with either the same root loss or similar insufficient root development. For instance, three of the 24 Bilos trees exhibited a limited taproot (270–350 mm long; corresponding to case 1); three trees presented none or very few windward roots (case 2); six trees presented insufficient perpendicular shallow roots (1–3 roots in total per tree, case 3); nine trees exhibited insufficient leeward roots (case 4); and sinker roots may also have been absent in windward or leeward sector (cases 5–7).

Tree anchorage strength predicted with simplified RSA

The critical turning moment T_0^{crit} predicted by RootAnchor for the reference root pattern (case 0) was 36.2 kN.m. This T_0^{crit} obtained from a simplified representation of the root system was of the same order as the empirical critical turning moments predicted by Cucchi et al. (2004) from trees of similar size ($T_{0,Cucchi}^{crit}$ between 47.1 and 59.5 kN.m, Eqs. (7)–(9)), but T_0^{crit} was 32 % lower than the $T_{0,Cucchi}^{crit}$ midpoint value (53.3 kN.m).

The simulated response curves of the tree turning moment T_i as a function of the deflection angle θ_i (Fig. 4 and Online Resource 8) exhibited a similar trend as previous experimental curves reported for a number of plant species and sizes (e.g., Fraser and Gardiner 1967; Ennos et al. 1993; Crook et al. 1997; Crook and Ennos 1998; Cucchi 2004; Lundström et al. 2007a; Lundström et al. 2007b; Yang et al. 2014). In all root patterns, T_i increased first with θ_i, with small drops or recoveries. Then, after reaching its critical value, T_i either stabilised at a residual turning moment or began to fall progressively. The predicted critical deflection angles θ_i^{crit} were generally larger than the angles measured in Cucchi (2004), about 12–20° in simulations (Online Resource 8) against 4–12° obtained from the response curves of Cucchi's experiments for trees of similar size.

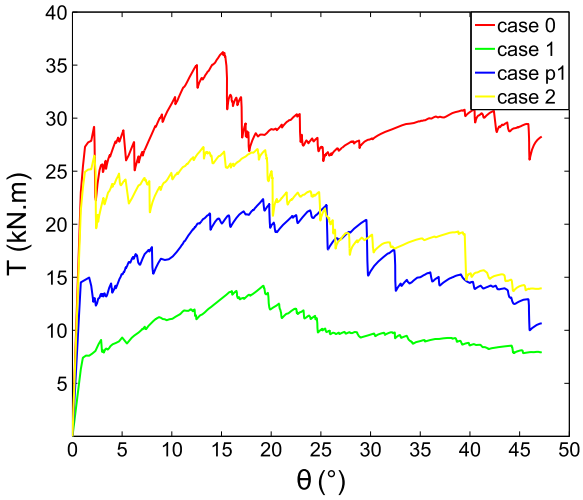


Fig. 4 Response curves of tree turning moment as a function of the deflection angle at the stem base, simulated by RootAnchor for cases 0, 1, 2 and p1. Other root system cases are presented in Online Resource 8

The tree overturning process

To examine how roots (i.e. stress types and failure pattern) may explain the progressive stiffness degradation of the root-soil system and to ultimate anchorage failure, Fig. 5 presents the stresses developing in shallow roots at the beginning of the overturning process ($\theta_i = 2-4°$) for cases 0, 1, 2 and p1, while Fig. 6 shows the broken roots (in red) at θ_i^{crit} for the same cases. Other cases exhibited less contrasting behaviour and are, thus, only presented in Online Resource 9.

In case 0, the first root damage occurred under bending loads in the lower part of the taproot (0.73–0.83 m deep), leading to the first drop in T_0 and reducing the stiffness of the root-soil system ($\theta_0 = 2°$). Afterwards, most leeward shallow roots were subject to bending and compression, and windward shallow roots to tension (Fig. 5a). Perpendicular shallow roots appear to have been less stressed than the wind-oriented shallow roots. At this stage, although the soil within the cage had started to fail, it remained close to the root stump. Overall, we observed little effect of soil failure on the response of root-soil system. Then, leeward shallow roots failed successively under bending loads at the root base, leading to another sequence of drops in T_0 ($3° \le \theta_0 \le 6°$). Then, several perpendicular shallow roots snapped successively from the leeward to the windward sides of the root system. The tree anchorage failure was finally reached when windward shallow roots within the cage failed in tension at $\theta_0^{crit} = 15°$. Throughout this overturning process, many sinker roots also snapped, mainly from the leeward to the windward sides, but with less impact on the tree turning moment. Overall, the stiffness of the root-soil system was severely reduced by the failure of the taproot and windward shallow roots, the latter leading to anchorage failure.

Without taproot (case 1), both the tree turning moment and the root-soil system stiffness were significantly lower at the beginning of the overturning process ($0° \le \theta_1 \le 5°$) compared to all the other cases. This led to a lower T_1^{crit} afterwards. The failure of shallow roots occurred almost simultaneously between the windward and leeward sides of the root system, compared to the prior damage of the leeward shallow roots in the cases with taproot (particularly cases 0 and 3). The absence of a taproot appears to have delayed the damage of several leeward roots compared to case 0, especially the chuck carrier (the thickest leeward shallow root). This delay was consistent with the lower leeward compressive

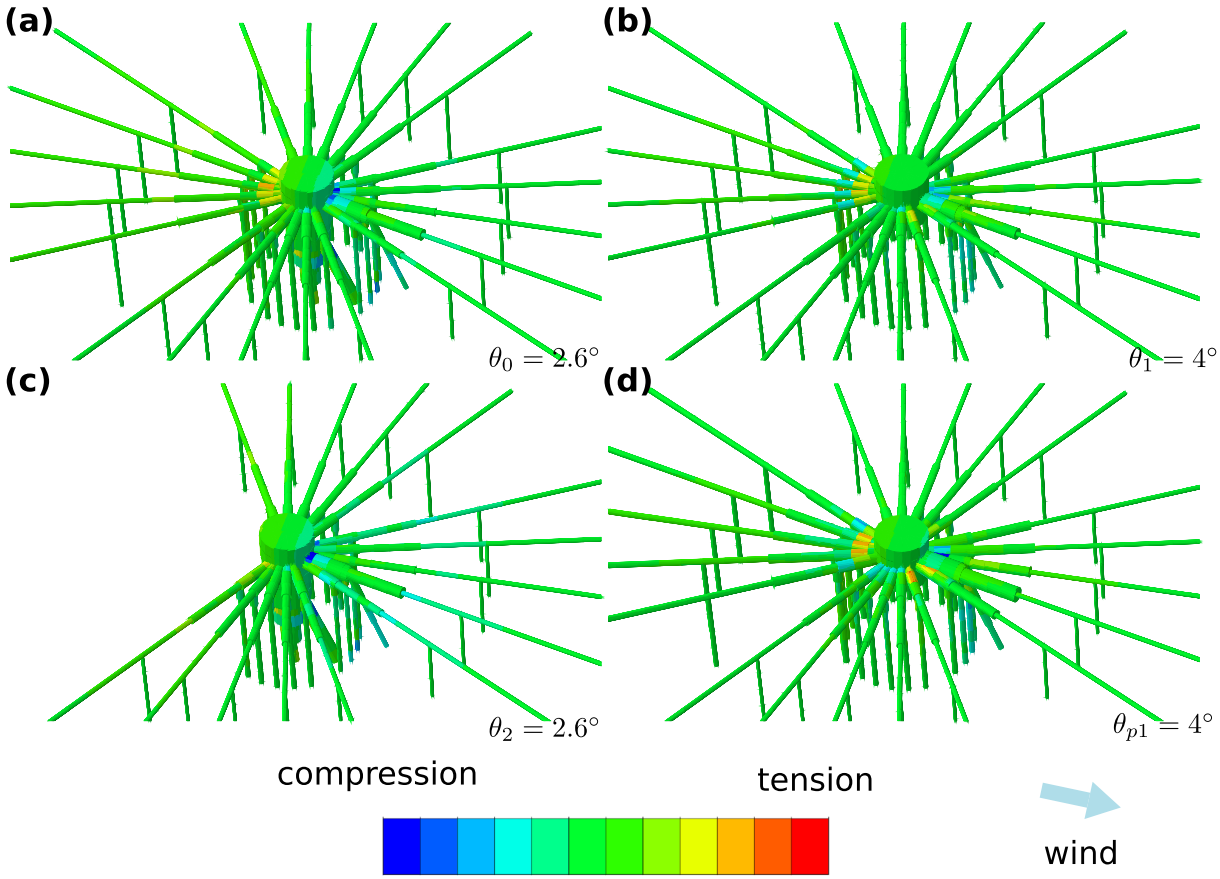


Fig. 5 Distribution of the axial stress intensity on root beams as simulated by RootAnchor at the beginning of the overturning process (2.6–4°) for 4 types of root patterns (cases 0–2, p1). Colours ranging from red to dark blue represent stress intensity from highest tension (positive) to highest compression (negative). (**a**) case 0 at deflection angle of 2.6°; (**b**) case 1 at deflection angle of 4°; (**c**) case 2 at deflection angle of 2.6°; (**d**) case p1 at deflection angle of 4.0°. Other root system cases are presented in Online Resource 9

stresses in case 1 than in case 0 developing at the beginning of tree overturning (Fig. 5).

With root secondary thickening in the leeward and windward sectors (case p1), both the turning moment and the initial root-soil system stiffness were improved compared to case 1 (Fig. 4). This led to T_{p1}^{crit} being higher than T_1^{crit}, but still weaker than in other cases. Stresses developing in windward and leeward shallow roots at the beginning of the process were similar to those in case 1. Windward shallow roots failed prior to any leeward shallow roots at $\theta_{p1} = 5°$, as opposed to the reference root pattern. The first damage in leeward shallow roots occurred at $\theta_{p1} = 10°$, but two leeward shallow roots appeared significantly loaded, failing after the turning moment reached its critical value ($\theta_{p1}^{\mathrm{crit}} = 19°$).

Without windward roots (case 2), the stresses developing in the taproot and leeward shallow roots at the beginning of the process and the root failure pattern were similar to case 0. The absence of windward roots led to two main differences compared to most of other cases: (1) a lower stiffness of the root-soil system after failure of the deep end taproot at $\theta_2 = 2°$ and (2) a lower T_2^{crit} at a smaller θ_2^{crit} (online Resource 8).

Contributions of the main structural roots to the tree anchorage strength

The critical turning moment T_i^{crit} simulated in each case is given in Table 3, along with the mechanical contribution C_i of each structural root class. The most significant variation in T^{crit} appeared to be driven by only a few root

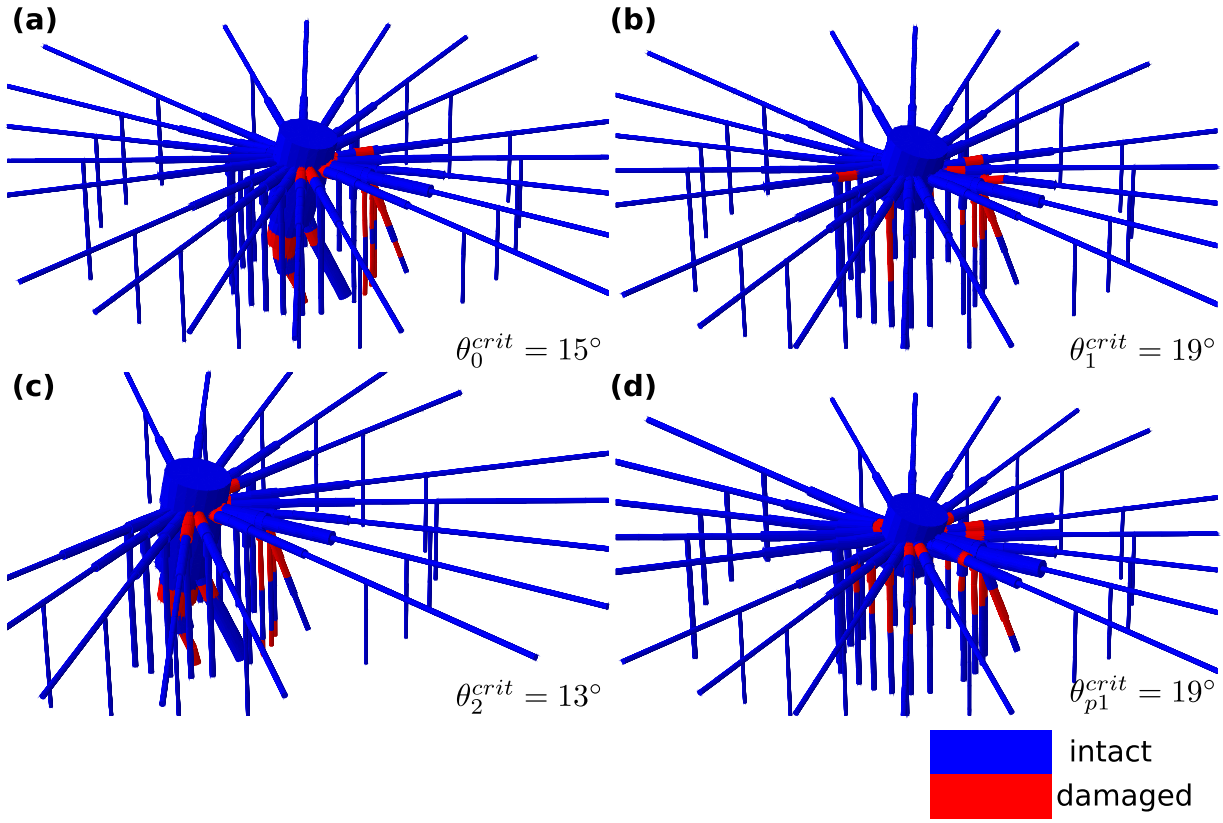


Fig. 6 Damage status of the root systems of cases 0–2 and p1 as simulated by RootAnchor at the critical moment (indicated by the *red colour*). Damage modes are described in the "Results" section. Other root system cases are presented in Online Resource 9

components. The taproot was the first mechanical contributor to tree anchorage strength with a decrease of 60.8 % in T_0^{crit} when the taproot is removed (case 1). The secondary root contributors were shallow roots and windward sinker roots: the critical turning moment was 24.8 % lower than T_0^{crit} with the loss of windward shallow roots, 20.2 % lower with the loss of perpendicular shallow roots, 11.6 % lower with the loss of windward sinker roots, and 6.9 % lower with the loss of leeward shallow roots. Some cases with minor root removals even predicted a slightly higher T^{crit} than the reference case. This difference (less than 2 %) remains in the uncertainty range of the critical turning moment predicted by the model, and thus cannot be attributed to a possible benefit of missing specific structural roots to the tree anchorage strength. Interestingly, with a stronger selective biomass reinforcement in wind-oriented sectors (case p1), 57.6 % of T_1^{crit} was recovered, leading to a reduced loss of 38.3 % of T_0^{crit}. This suggests that the wind-induced root adaptive plasticity (with full taproot biomass recovered) improved tree anchorage but not enough to compensate for the absence of a taproot.

Discussion

Our in silico analysis using a simplified RSA has predicted realistic tree anchorage strength and individual root response to tree overturning compared to tree-pulling experiments and empirical anchorage strength for *P. pinaster*. The model allowed us also to estimate the contribution of the main structural roots to the tree anchorage strength. We now examine these results with regards to previous experimental findings on anchorage mechanics. We also try to reconcile disparate findings obtained for different plant species and various soil conditions from tree-pulling experiments and root architecture analyses and then discuss improvement of tree anchorage by wind-induced root acclimation.

The main root architectural components in tree anchorage

The tree overturning process appeared to have well reproduced with a simplified root spatial arrangement. The simplified sinker root systems agreed with experimental observations of the taproot as a pole in the soil, guyed by large shallow roots extending symmetrically and radially outward from the stem base, and with windward and perpendicular sinker roots extending from the shallow roots (Danjon et al. 2005). With a simplified RSA (case 0), the simulated anchorage strength represented 68 % of the empirical value obtained by Cucchi et al. (2004) (Eqs. (7)–(9)) for *P. pinaster*. We believe that our model prediction could be further improved by increasing soil resistance within the rigid cage and by using the specific growth conditions of the tree.

Simulations showed that the contribution of the structural roots to tree anchorage depends on the mechanism of progressive degradation of the structural roots (Figs. 4 and 6). Additionally, the role of structural roots differs depending on their orientation, morphology, and changes during the root system degradation (Figs. 5 and 6). The taproot first bends. After its failure, the windward shallow roots undergo tension further from the stem base, and the leeward shallow roots undergo compression, and fail close to the root base. This pattern agrees well with most field observations (Table 4). In particular, our simulations highlight: (1) the ultimate anchorage failure occurring at the tensile failure of the windward shallow roots, and (2) the hinge support provided by leeward shallow roots at the beginning of tree overturning. However, we observed little effect of soil failure on the root-soil system response, which differs from the prediction of Coutts (1986) for mature trees. Our model probably needs further improvement in soil resistance modelling.

Our model predicted a dominant role of the taproot in tree anchorage. The taproot accounted for 60.8 % of the anchorage strength in case 0 and overall provided a high rigidity to the root-soil system (Fig. 4). This dominant role is consistent with previous experimental findings for relatively smaller trees (Crook and Ennos 1998; Danjon et al. 1999b). Our result supports also (1) a recent study on the protective effects of a large and deep taproot in young *P. pinaster* trees against leaning and toppling (Danquechin Dorval et al. 2016), and (2) a previous numerical study with 2D root patterns on a sandy soil showing a contribution of the taproot on anchorage strength greater than 50 % (Fourcaud et al. 2008). However, this dominant role of the taproot should be true only for young trees. As sinker RSA trees grow, the influence of the taproot on the tree anchorage usually diminishes, the taproot stops its growth and the central part of the root-soil system forms a rigid cage in which roots and soil act together as an entity (Danjon et al. 2005).

Our simulations confirm the importance of a symmetric circular distribution of the shallow roots for sinker root systems. First, windward shallow roots are the second most important root component after the taproot in contributing to anchorage strength (24.8 %). These roots are stressed in tension at the critical moment, and their failure leads to the ultimate failure of the root-soil system. Second, perpendicular shallow roots are non-negligible as they relay the response of the root-soil system after the failure of leeward roots and before the active response of windward roots. This intermediate role leads to a significant contribution to the anchorage strength (20.2 %). Third, the leeward roots, along with the taproot, provide the anchorage rigidity at the beginning of tree overturning, although their contribution to anchorage strength represents only 6.9 %. The importance of this radial distribution of shallow roots was also suggested by root architecture studies (Coutts et al. 1999; Danjon et al. 2005); while studies based on tree-pulling experiments focused only on the roles of windward and leeward shallow roots (Coutts 1983; Coutts 1986; Crook and Ennos 1996; Crook et al. 1997; Stokes 1999).

Toward a better understanding of the impact of wind-induced acclimation on tree anchorage

RootAnchor was successful in simulating the intensity and distribution of mechanical stresses, and the resulting damage to a root-soil system under wind load. Since wind-induced acclimation depends on the distribution of external mechanical loads on plant elements over the constitutive plant tissue (Coutand et al. 2008; Bonnesoeur et al. 2016), RootAnchor can provide an analysis of the mechanical importance of the root acclimation.

A first interesting application of RootAnchor was to predict the location of root element acclimation within the root system by assuming that root elements acclimate in high strain spots. Following this assumption, our simulations lead to a preferential acclimation in the

Table 4 Experimental evidence of the role of RSA in tree overturning reported in the literature

Root types	Response to overturning	Species observed	References
Root-soil system	Basal area of the root system are most stressed and eventually failed mechanically in some cases	*Helianthus annuus; P. pinaster; Larix europea x L. japonica*	Ennos 1989; Crook and Ennos 1996; Stokes 1999
	A root-soil plate may be formed and lifted by overturning for mature root systems	*P. sitchensis; P. pinaster; Samanea saman*	Coutts 1983; Coutts 1986; Ray and Nicoll 1998; Stokes et al. 1998; Danjon et al. 2005; Lundström et al. 2007a; Ow et al. 2010
Taproot	Bent or rotated in the soil; eventually failed close to the base	*P. pinaster; Larix europea x L. japonica; Mallotus wrayi*	Ennos et al. 1993; Crook and Ennos 1996; Crook et al. 1997; Stokes 1999
	Most important component for young trees; its importance depending also on its dimensions relative to the size of the aerial parts	*Larix europea x L. japonica; P. pinaster; Mallotus wrayi*	Crook and Ennos 1996; Crook and Ennos 1998; Danjon et al. 2005; Khuder et al. 2007
Windward laterals along with sinker roots	Moved upwards; stressed in tension and eventually in shear; likely to fail at the moment when T^{crit} was reached, either at the base or further from the stem base, especially at the junction with sinker roots; could be lifted up	*Larix europea x L. japonica; P. sitchensis; P. pinaster; Mallotus wray*	Coutts 1983; Coutts 1986; Crook and Ennos 1996; Crook et al. 1997; Crook and Ennos 1998; Ray and Nicoll 1998; Stokes 1999
	The most important component to root anchorage for shallowly rooted systems; generally important to root anchorage	*Larix europea x L. japonica; P. sitchensis*	Coutts 1986; Ennos et al. 1993; Crook and Ennos 1996
Leeward roots	Pushed deeper in soil; stressed in compression & bending; likely to fail at the root base; for plate-like root systems they were likely to fail after the anchorage failure had been reached	*P. sitchensis; Larix europea x L. japonica; Mallotus wrayi; P. pinaster*	Coutts 1983; Coutts 1986; Crook and Ennos 1996; Crook et al. 1997; Stokes 1999
	Less important than windward roots for shallowly rooted systems	*P. sitchensis; Impatiens glandulifera; H. annuus*	Coutts 1986; Ennos et al. 1993
Small roots	Likely to break throughout the overturning process, particularly from leeward side toward the windward side in some cases	*P. sitchensis; Larix europea x L. japonica*	Coutts 1983; Coutts 1986; Crook and Ennos 1996

middle of the taproot and on windward and leeward shallow roots within the cage where stresses were higher. On the other hand, roots growing in the sectors perpendicular to the wind should be less concerned by acclimation (Fig. 5). This prediction is in agreement with numerous root acclimation observations (Stokes et al. 1995; Nicoll and Ray 1996; Stokes et al. 1997b; Danjon et al. 2005).

A second interesting application of RootAnchor is to quantify the impacts of wind-induced root acclimation on tree response to uprooting. The comparison between cases p1 and 1 highlighted the mechanical importance of leeward roots. The leeward hinge support with adaptive secondary thickening enhances tree anchorage by providing an higher initial rigidity of the root-soil system (see initial slopes of cases 0 and p1 in Fig. 4: Coutts 1983; Coutts 1986; Nicoll and Ray 1996). This benefit of leeward roots acclimation on tree anchorage was suggested by root architecture analyses (Nicoll and Ray 1996; Danjon et al. 2005). In the literature, the role of the leeward roots in tree anchorage remains unclear. Some tree-pulling experiments predicted a minor contribution

of leeward roots (Coutts 1986) whereas root architecture analyses showed pronounced adaptive growths of leeward roots due to winds, which should increase the mechanical support of leeward roots to the tree anchorage (Nicoll and Ray 1996). This difference between tree-pulling experiments and root architecture analyses could be explained by the underestimation by tree-pulling experiments of the leeward contribution by pulling the tree in a direction other than the prevailing wind. Additionally, the leeward hinge support in tree anchorage should be sensitive to the diameter of the leeward roots as the root bending stiffness and the maximum bending stress increases with the fourth and third power of the root diameter, respectively (Timoshenko 1930). In contrast, the tensile force to break windward shallow roots increases only with the square of the root diameter, leading to weaker effects than leeward roots. As a consequence, a small difference in leeward root diameter may lead to visible changes in its contribution to tree anchorage.

Although wind-induced root acclimation has been reported for different species, little is known about its direct impacts on tree response to uprooting (Stokes et al. 1995; Tamasi et al. 2005; Danjon et al. 2013b). Our first attempt to link root acclimation of *P. pinaster* to the anchorage function captures the compensatory mechanism of local root secondary thickening. Further investigations should examine the mechanical impacts of various forms of root acclimation on the root anchorage function by adapting root architectural and material parameters (root number, length, diameter, shapes of cross-sections, wood density, modulus of elasticity and modulus of rupture, etc.).

Conclusion

By using a 3D biomechanical model, we demonstrate the possibility of simplifying the tree root system architecture (RSA) for modelling tree anchorage, and find that only a few components of sinker RSA are responsible for below-ground anchorage of tap-rooted coniferous species growing in sandy podzols. These components are classified as follows in order of their quantitative importance to tree anchorage strength: taproot > windward shallow roots > perpendicular shallow roots > windward sinker roots > leeward shallow roots.

Simulations also suggest that the most wind-acclimated roots correspond to the highest stressed roots at the early loading stage. Anchorage is expected to be more sensitive to acclimation of the leeward hinge roots by adaptive secondary thickening than the windward roots.

More generally, this study demonstrates that in silico analysis is a way (1) to provide a realistic picture of root stress and failure patterns during tree overturning, (2) to determine the role of each tree anchorage component with the confounding factors under control, (3) to analyse the stresses that a tree root system undergoes at specified locations, and (4) to quantify the stability benefits of a tree by adapting its biomass allocation to roots. The approach developed in this study could be further applied to other species growing under various conditions by updating root and soil properties. This indicates the potential of our model for improving tree anchorage in risk models predicting windstorm damage (Peltola et al. 1999; Gardiner et al. 2000; Hale et al. 2015; Dupont 2016).

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